

Sale Price, Sales Tax, and Finding the Original Amount

Student: Isaac Lee

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Standard: CCSS.MATH.CONTENT.7.RP.A.3 — Discounts, sales tax, finding the whole given a part and the percent.

Part A — Find the Sale Price

Problem 1

A jacket originally costs \$90 and is on sale for 30% off. Find the sale price.

Discount

Discount: $0.30 \times \$90 = \27 .

Sale price: $\$90 - \$27 = \$63$.

\$63

Problem 2

A pair of boots that costs \$120 has a 35% off sign. Find the sale price.

Discount

Discount: $0.35 \times \$120 = \42 .

Sale price: $\$120 - \$42 = \$78$.

\$78

Problem 3

A laptop priced at \$800 is on sale for 15% off. Find the sale price.

Discount

Larger Base

Discount: $0.15 \times \$800 = \120 .

Sale price: $\$800 - \$120 = \$680$.

\$680

Problem 4

A bicycle that normally costs \$250 is marked down 60%. Find the sale price.

Discount

Discount: $0.60 \times \$250 = \150 .

Sale price: $\$250 - \$150 = \$100$.

\$100

Part B — Find the Total Cost with Sales Tax

Problem 5

A board game costs \$40, and there is a 5% sales tax. Find the total cost.

Sales Tax

Tax: $0.05 \times \$40 = \2 .

Total: $\$40 + \$2 = \$42$.

\$42

Problem 6

A pair of shoes priced at \$75 has an 8% sales tax. Find the total cost.

Sales Tax

Tax: $0.08 \times \$75 = \6 .

Total: $\$75 + \$6 = \$81$.

\$81

Problem 7

A water bottle costs \$30 and has a 10% sales tax. Find the total cost.

Sales Tax

Tax: $0.10 \times \$30 = \3 .

Total: $\$30 + \$3 = \$33$.

\$33

Problem 8

A camera priced at \$200 has a 7% sales tax. Find the total cost.

Sales Tax

Tax: $0.07 \times \$200 = \14 .

Total: $\$200 + \$14 = \$214$.

\$214

Part C — Find the Original Price from the Sale Price

Problem 9

After 25% off, a shirt costs \$60. What was the original price?

Reverse

75% remains

After 25% off, you pay $100\% - 25\% = 75\%$ of the original.

Equation: $0.75 \times x = 60$.

$$x = \frac{60}{0.75} = \$80.$$

Check: 25% of 80 = 20; $80 - 20 = \$60$ ✓.

\$80

Problem 10

After 20% off, a watch sells for \$40. What was the original price?

Reverse

80% remains

After 20% off, sale = 80% of original. Equation: $0.80 \times x = 40$.

$$x = \frac{40}{0.80} = \$50.$$

Check: 20% of 50 = 10; $50 - 10 = \$40$ ✓.

\$50

Problem 11

After 50% off, a backpack costs \$35. What was the original price?

Reverse

Half-price

After 50% off, sale = 50% of original (half).

Equation: $0.50 \times x = 35 \Rightarrow x = 35 \times 2 = \70 .

\$70

Problem 12

A book is on sale for 10% off and now costs \$27. What was the original price?

Reverse

90% remains

After 10% off, sale = 90% of original. Equation: $0.90 \times x = 27$.

$$x = \frac{27}{0.90} = \$30.$$

\$30

Part D — Find the Original Price from the Total Cost

Problem 13

Including a 5% sales tax, a meal costs \$63. What was the price before tax?

Reverse Tax

105%

Total = original + 5% tax = 105% of original.

Equation: $1.05 \times x = 63$.

$$x = \frac{63}{1.05} = \$60.$$

Check: 5% of 60 = 3; 60 + 3 = \$63 ✓.

\$60

Problem 14

Including a 10% sales tax, a hat costs \$44. What was the price before tax?

Reverse Tax

110%

Total = 110% of original. Equation: $1.10 \times x = 44$.

$$x = \frac{44}{1.10} = \$40.$$

\$40

Problem 15

Including a 4% sales tax, a notebook costs \$26. What was the price before tax?

Reverse Tax

104%

Equation: $1.04 \times x = 26$.

$$x = \frac{26}{1.04} = \$25.$$

\$25

Problem 16

Including a 20% sales tax, a model car costs \$48. What was the price before tax?

Reverse Tax

120%

Equation: $1.20 \times x = 48$.

$$x = \frac{48}{1.20} = \$40.$$

\$40

Part E — Find the Rate

Problem 17

A toy was originally 80 and is now on sale for 60. What is the discount rate (% off)?

Find Rate

Discount

Discount amount: $\$80 - \$60 = \$20$.

Rate: $\frac{20}{80} = 0.25 = 25\%$.

25%

Problem 18

A pen was originally priced at 20, and the total cost with sales tax was 21. What is the sales tax rate?

Find Rate

Tax

Tax amount: $\$21 - \$20 = \$1$.

Rate: $\frac{1}{20} = 0.05 = 5\%$.

5%

Part F — Multi-Step (Discount, Then Tax)

Problem 19

A sweater is originally priced at \$80 and is on sale for 25% off. After the discount, a 10% sales tax is applied. Find the final total cost.

Multi-Step

Discount → Tax

Step 1 — Discount: $0.25 \times \$80 = \20 . Sale price: $\$80 - \$20 = \$60$.

Step 2 — Tax on sale price: $0.10 \times \$60 = \6 .

Step 3 — Final total: $\$60 + \$6 = \$66$.

\$66

Problem 20

A skateboard regularly costs \$100. The store offers 40% off, then adds a 5% sales tax to the discounted price. Find the final total cost.

Multi-Step

Discount → Tax

Step 1 — Discount: $0.40 \times \$100 = \40 . Sale price: $\$100 - \$40 = \$60$.

Step 2 — Tax on sale price: $0.05 \times \$60 = \3 .

Step 3 — Final total: $\$60 + \$3 = \$63$.

\$63